CSP – Logic

* Find $\underline{\underline{w}}$ / spatial filter such that : [$w^{n \times 1}$ where n = num chans]

$$\max \left(\frac{\mathrm{var}(w^T X_{\text{class } A})}{\mathrm{var}(w^T X_{\text{class } A}) + \mathrm{var}(w^T X_{\text{class } B})} \right)$$

→ so increase variance of one class and decrease the variance of the other

what is $w^T X$? $\Rightarrow S(t)$ ∴ X is EEG data

$$\Rightarrow \left(w_{N \times 1}\right)^T \cdot \underset{t \times N \times \text{Samp}}{X} = S$$

t ↓ trials, N ↓ chans, Samp ↓ samples

$$\Rightarrow w_{1 \times N} \cdot X_{N \times t \times \text{samp}} = S_{1 \times t \times \text{samp}}$$

so S is a vector with num trials × samples time series.

$$S = \begin{bmatrix} \cdots \\ \cdots \\ \cdots \end{bmatrix}_{t \times \text{sam}} \; : \; \mathrm{var}(S) = \begin{bmatrix} \vdots \end{bmatrix}_{t \times 1}$$

CSP_ Math

So, we understood how $var(w^T X)$ is computed.
But how do we actually maximize $\dfrac{var(w^T X_A)}{var(w^T X_A) + var(w^T X_B)}$?

A, B ⇒ classes

if $L = \dfrac{var(w^T X_A)}{var(w^T X_A) + var(w^T X_B)}$, then we do:

$$\frac{dL}{dw} = 0$$

Mathematically, $var(w^T X_k) = w^T \bar{C}_k w$ where C_k is the covariance matrix of that class. ($Covar = \dfrac{X \cdot X^T}{N-1}$)

How is $var(w^T X_k) = w^T \bar{C}_k w$? Let's find out.

Assuming $\underset{T\times 1}{S}$ ($S = w^T X_k$) is mean centered ($S_{mean} = 0$)
↓
EEG is mean centered)

$$var(S) = \frac{1}{T-1} \sum_t (S(t))^2 = \frac{1}{T-1} \sum_t (w^T . X_k)^2$$

$$\Rightarrow \frac{1}{T-1} \sum_t (w^T . X_k)^2 \Rightarrow \frac{1}{T-1} (w^T X_k) \underbrace{(w^T X_k)^T}_{X_k^T w}$$

$$\Rightarrow \frac{1}{T-1} (w^T \underbrace{X_k X_k^T}_{\bar{C}_k = \frac{X_k . X_k^T}{T-1}} w)$$

CSP Math

$var(S) = w^T \bar{C}_A w$, now our ratio becomes

$$L = \max \left[\frac{(w^T \bar{C}_A w)}{w^T C_A w + w^T C_B w} \right] = \frac{w^T \bar{C}_A w}{w^T (\bar{C}_A + \bar{C}_B) w}$$

$\frac{dL}{dw} = 0$, this gives us: don't ask how :)

$$\boxed{\bar{C}_A w = \lambda (\bar{C}_A + \bar{C}_B) w}$$

↳ once w is computed

$$\lambda_i = \frac{\bar{C}_A w}{(\bar{C}_A + \bar{C}_B) w}$$

⇒ for C $N \times N$, we get a total of $N \times N$ dimensional

W:

⇒ W ⇒ $W_{N \times N}$, with each W_i corresponding

to a λ_i - $\lambda = \begin{bmatrix} \vdots \end{bmatrix}_{N \times 1}$. w_i's are picked accordingly!!

★ If λ close to 0 → variance of class B is maximized

" " " 1 → " " class A " "